Integrating Fasting expression data to the EMT SBML model*

Integrating Fasting-Induced Gene Expressions with EMT Signaling Networks

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Abstract

Contains integrated fasting-induced gene expression data with the epithelial-mesenchymal transition (EMT) signaling networks using computational methods. The EMT model, derived from SBML pathway maps and enhanced with COBRApy, was refined to incorporate gene regulatory relationships and metabolic constraints. Focusing on the impact of fasting on EMT metastasis, we identified key genes and reactions influenced by differential gene expression profiles. Through relaxation techniques and flux variability analysis, we simulated the metabolic responses under fasting conditions, revealing potential regulatory nodes and metabolic pathways affected during EMT. Our approach provides insights into how fasting-induced gene expression changes may modulate EMT dynamics, offering a computational framework for studying metabolic adaptations in cancer progression.

```python
import cobra
import pandas as pd
```

Load EMT Model and the Expression data

```python
# Load the SBML model
model = cobra.io.read_sbml_model(
    "/Users/karthik/Desktop/PHCCO IISc Internship/EMT/MODEL1602080000_url.xml")

# Load the expression data
expression_data = pd.read_excel(
    "/Users/karthik/Desktop/PHCCO IISc Internship/EMT/example.xlsx",
    sheet_name="Sheet1", header=None)
```

**

Specify the column names in a sequence

```
expression_data.columns = ['uniprot_id', 'expression_data', 'p_value']
```

Convert to expression data to dict for easy handling

```
# Convert the expression data to a dictionary
expression_dict = expression_data.set_index('uniprot_id')['expression_data'].to_dict()
expression_dict
```

```
{'P60484': 4.6975102,
 'Q92934': 4.477713,
 'P28062': 4.448816,
 'P06241': 4.3806553,
 'O60674': 4.317239,
 'P27361': 4.2146053,
 'Q05655': 4.1526284,
 'Q6IAA8': 4.1215506,
 'P29353': -2.2021408,
 'P18433': -2.4953408,
 'Q5T1C6': -2.6014206,
 'Q8NDV7': -2.614702}
```

Integrate expression

Returns all the affected up and down reactions, using a predefined threshold, as tuple of lists of the each.

```
# Define function to integrate expression data
def integrate_expression(model, expression_dict, up_threshold=2, down_threshold=-2):
    up_genes = [gene for gene, expr in expression_dict.items() if expr >= up_threshold]
    down_genes = [gene for gene, expr in expression_dict.items() if expr <= down_threshold]

    affected_up_rxns = []
    affected_down_rxns = []

    for rxn in model.reactions:
        genes = [g.id for g in rxn.genes]
        if any(gene in up_genes for gene in genes):
            affected_up_rxns.append(rxn.id)
        if any(gene in down_genes for gene in genes):
            affected_down_rxns.append(rxn.id)

    return affected_up_rxns, affected_down_rxns

# Identify affected reactions
up_rxns, down_rxns = integrate_expression(model, expression_dict)
```

Adjust bounds

Modified upper and lower bounds of reactions based on the up and down affected reactions list above. They are scaled by `factor`

```
# Define function to adjust bounds
def adjust_bounds(model, up_rxns, down_rxns, factor=0.01):
    for rxn_id in up_rxns:
        rxn = model.reactions.get_by_id(rxn_id)
        rxn.upper_bound *= factor
        rxn.lower_bound *= factor

    for rxn_id in down_rxns:
        rxn = model.reactions.get_by_id(rxn_id)
        rxn.upper_bound *= factor
        rxn.lower_bound *= factor

# Adjust reaction bounds based on expression data
adjust_bounds(model, up_rxns, down_rxns)
```

Save the integrated model

```
# Save the modified model
cobra.io.write_sbml_model(model, 'integrated_MODEL1602080000.xml')
```